

Enterprise Content Intelligence (PoC)

Healthcare Content Lifecycle Management using AI

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Problem Statement & Challenges

Healthcare organizations manage : Billing policies, Clinical practice guidelines, Provider contracts, Coding manuals, Frequent policy updates

Challenges

- Manual review
- Slow onboarding
- Compliance risks
- Financial risk
- Difficult policy comparison

Mission: Automate ingestion, audit policy changes, and seamlessly compile prose descriptions into executable programming code

Industry Trends

- Retrieval-Augmented Generation
- Local Large Language Models
- Semantic Search
- Vector Databases
- Structured Knowledge Extraction
- Policy Automation

Proposed Solution & PoC

- Proposed solution : Automated healthcare content intelligence system that Ingests and compares healthcare documents to validate operational changes and convert narrative policies into executable code

Three workflows/tracks of PoC:

- document ingestion, semantic indexing, RAG, structured information extraction with schema validation (Intelligent Ingestion & Verified RAG)
- compares two versions of healthcare policies & identifies important operational differences (Payer-Provider Policy Shift Audit)
- converts narrative healthcare policies into executable Python functions and validates those functions using sample patient data (Policy-to-Code Transformation)

Demonstration

- PDF Upload
- Semantic Search
- RAG Question Answering
- Structured JSON
- Policy Comparison
- Policy-to-Code
- Rule Validation

Technology Stack

- Frontend : Streamlit
- AI/LLM : Ollama, Llama 3.1
- NLP : Sentence Transformers
- Database : ChromaDB
- Parsing : PyMuPDF
- Validation : Pydantic

Business Value

- Faster policy review
- Better consistency
- Reduced manual effort
- Improved compliance
- Privacy-first AI
- Scalable architecture

Strategic Recommendations & Growth Opportunities

- Instant Rule Deployment
- Automated Policy Comparison
- On-Premise AI Assistants
- Enterprise Knowledge Platform

Future Enhancements & Conclusion

- Modern AI for Healthcare Content
- Privacy-Preserving Local Architecture
- Working Proof of Concept
- Future Enhancement Opportunities

Thank you!